Course Name	The Theory of Linear Programming (Midterm)	Exam Duration		Exam Date	25/11/2021
Course Instructor	Assoc. Prof. Dr. Hale Gonce Kocken		Signature		
Student Disciplinary Regulations "and to make or attempt to make copies of exams to" the actual perpetrators are suspended from one or two semesters. (YÖK; 2547 Student Disciplinary Regulations, 9. Article)					

Q1. (15 pts) Select the correct answer - true or false.

i. The situation that occurs when we have no solution that satisfies the constraints is referred to as infeasibility. T F

ii. When the optimal solution is found, all slack and surplus variables have a value of zero. T F

iii. The Z row in the simplex table tells us whether the current solution is optimal, and, if it is not, what variable will be in the optimal solution. T F

iv. In any linear programming problem, if a variable is to enter the solution, it must have a positive coefficient in the Z row. T F

v. If all the numbers in the pivot column are positive, this implies that the solution is unbounded. T F

Q2. (5 pts) Simplex method of solving linear programming problem uses

a) all the points in the feasible region
b) only the corner points of the feasible region
c) intermediate points within the infeasible region
d) only the interior points in the feasible region

Q3. (5 pts) If x is a decision variable of a linear programming problem and unrestricted in sign then this variable can be converted into $x = x' - x''$ so as to solve the problem by simplex method, where:

a) $x' \le 0$ and $x'' \ge 0$ b) x' and $x'' \le 0$

c) $x' \ge 0$ and $x'' \le 0$ d) x' and $x'' \ge 0$

Q4) (5 pts) While solving a linear programming problem by simplex method, if all pivot column coefficients are negative, then the problem has which of the following types of solution?

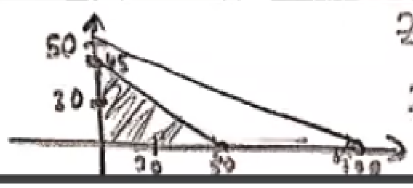
a) An unbounded solution b) Alternative solutions c) A unique solution d) No solution

Q5) (10 pts) One unit of product P_1 requires 3 kg of resource R_1 and 1 kg of resource R_2. One unit of product P_2 requires 2 kg of resource R_1 and 2 kg of resource R_2. The profits per unit by selling product P_1 and P_2 are \$2and \$3, respectively. The manufacturer has 90 kg of resource R_1 and 100 kg of resource R_2. The manufacturer can make a maximum profit of ....

Hint: You may use the graphical solution method.

a) [redacted] 100	b) 135	c) 150	d) 200

$(0,45) \leftarrow 3x_1 + 2x_2 \le 90$ $(0,50)$
$(30,0)$ $\nearrow (100,0)$
$x_1 + 2x_2 \le 100$
$x_1, x_2 \ge 0$
max $z = 2x_1 + 3x_2$



$z_{(0,45)} = 135$

$z_{(50,0)} = 100$

I wish you success.

Q6) (10 pts)

Starting tableaux

Basic	x_1	x_2	x_3	x_4	x_5	Solution
z	-1	a $-\frac{1}{2}$	-1	b 0	c 0	0
X_4	2	d 1	1	e 1	0	f = 12
X_5	1	3	g 2	h 0	1	24

Second tableaux

Basic	x_1	x_2	x_3	x_4	x_5	Solution
z	i 0	-1/2	j $-\frac{1}{2}$	k $\frac{1}{2}$	0	m = 6
	1	1/2	n $\frac{1}{2}$	1/2	0	6
	0	p $\frac{5}{2}$	3/2	-1/2	1	r = 18

$1 \quad \frac{1}{2} \quad \frac{1}{2} \quad \frac{1}{2} \quad 0 \quad 6$

$-1 \quad a \quad -1 \quad 0 \quad 0 \quad 0$

$0 \quad -\frac{1}{2} \quad j \quad k \quad 0 \quad m$

$-1 \quad -\frac{1}{2} \quad -\frac{1}{2} \quad -\frac{1}{2} \quad 0 \quad -6$

$+\;1 \quad 3 \quad g \quad 0 \quad 1 \quad 24$

$0 \quad p \quad \frac{3}{2} \quad -\frac{1}{2} \quad 1 \quad r$

$\cancel{-1}$, $\cancel{1}$, 1, 12, 2, 15, $\cancel{-1/2}$, $\cancel{1/2}$, 6, 6, 1/2, 5/2, 3, 9, + 18, 18, 27, 15, 42

The first and the second tableau of a linear programming problem are given above. Find the summation of all unknowns, that is a+b+c+d+e+d+e+f+g+h+i+j+k+l+m+n+p+r = ?

a) 38	b) 38.5	c) 42	d) 44.5

Q7) (15 pts) A company manufactures three products (x_1, x_2, x_3) whose unit profits are \$2, \$5, and \$3, respectively. The company has budgeted 80 hours of labor time and 65 hours of machine time for the production of three products. The labor requirements per unit of products 1, 2, and 3 are 2, 1, and 2 hours, respectively. The corresponding machine-time requirements per unit are 1, 1, and 2 hours. The company regards the budgeted labor and machine hours as goals that may be exceeded, if necessary, but at the additional cost \$15 per labor hour and \$10 per machine hour. Formulate the problem as an LP.

$$\boxed{\max z = 2x_1 + 5x_2 + 3x_3 - 15x_4'' - 10x_5''}$$

$$2x_1 + x_2 + 2x_3 \le 80 \rightarrow \boxed{2x_1 + x_2 + 2x_3 + x_4' - x_4'' = 80} \qquad x_4 = x_4' - x_4'' \text{ free}$$

$$x_1 + x_2 + 2x_3 \le 65 \rightarrow \boxed{x_1 + x_2 + 2x_3 + x_5' - x_5'' = 65} \qquad x_5 = x_5' - x_5'' \text{ free}$$

$$\boxed{x_1, x_2, x_3, x_4', x_4'', x_5', x_5'' \ge 0}$$

Q8.

$\min z = x_1 + 2x_2$

s.t. $x_1 + x_2 \geq 2$

$3x_1 + 4x_2 \leq 8$

$x_1 \geq 0, x_2 \geq 0$

a) (10 pts) Solve the given problem by algebraic method and write the optimal solution.

b) (15 pts) Solve the given problem by Simplex method and write the optimal solution.

$\min z = x_1 + 2x_2$

$x_1 + x_2 - x_3 = 2$

$3x_1 + 4x_2 + x_4 = 8$

$x_1, x_2, x_3, x_4 \geq 0$

$\binom{4}{2} = \frac{4!}{2!2!} = \frac{\cancel{4} \cdot 3}{\cancel{2}} = 6.$

(1) $x_1 = 0,\ x_2 = 0,\ x_3 = -2,\ x_4 = 8 \quad z = -$

(2) $x_1 = 0,\ x_3 = 0,\ x_2 = 2,\ x_4 = 0 \quad z = 4$

(3) $x_1 = 0,\ x_4 = 0,\ x_2 = 2 \quad x_3 = 0 \quad z = 4$

(4) $x_2 = 0,\ x_3 = 0,\ x_1 = 2,\ x_4 = 2. \quad z = 2$ *

(5) $x_2 = 0,\ x_4 = 0,\ x_1 = 8/3,\ x_3 = 2/3 \quad z = \frac{8}{3} = 2,..$

(6) $x_3 = 0,\ x_4 = 0,\ x_2 = 2,\ x_1 = 0 \quad z = 4$

$-3x_1 - 3x_2 = -6$

$3x_1 + 4x_2 = 8$

$x_2 = 2$

$x_1^* = 2,\ x_2^* = 0$

$x_3^* = 0 \quad x_4^* = 2$

$z^* = 2$

optimal.

max (−)

min (+)

$\min z = x_1 + 2x_2 + MR$

$x_1 + x_2 - x_3 + R = 2$

$3x_1 + 4x_2 + x_4 = 8$

$x_1, x_2, x_3, x_4, R \geq 0$

$R = 2 - x_1 - x_2 + x_3$

$z = x_1 + 2x_2 + M(2 - x_1 - x_2 + x_3)$

$= (1-M)x_1 + (2-M)x_2 + Mx_3 + 2M$

Basic	x_1	x_2	x_3	x_4	R	RHS	
z	M-1	M-2	-M	0	0	2M	
R	1	1	-1	0	1	2	2 →
x_4	3	4	0	1	0	8	8/3
z	0	-1	-1	0	1-M	2	
x_1	1	1	-1	0	1	2	
x_4	0	1	3	1	-3	2	

$x_1^* = 2 \quad x_2^* = 0 \quad x_3^* = 0 \quad x_4^* = 2$

$z^* = 2$

I wish you success.